

UPE Functional Blocks Architecture - Skeleton Framework

What UPE Must Be Capable of Doing

Document Version: 1.0 (Draft Skeleton)

Purpose: Define the functional boundaries and capability domains of the Unified Project Execution platform at a high level, creating a pool of functional blocks to be detailed and prioritized in subsequent phases.

Methodology Note: This document extracts functional capabilities from all project conversations, meeting transcripts, and brainstorming sessions. Duplicate, contradictory, and complementary functions are included intentionally to provide a comprehensive pool for later evaluation and prioritization.

SECTION 1: PROJECT LIFECYCLE & ENVIRONMENT MANAGEMENT

1.1 Project Initialization & Provisioning

Core Capability: Automated project setup from inception to operational state

- **Project Creation Interface**
 - Ability to select project type from portfolio/taxonomy
 - Ability to define project class/category (drives template selection)
 - Ability to input basic project metadata (name, code, location, client, disciplines)
 - Ability to define project scope and boundaries
 - Ability to assign initial project stakeholders/roles
 - Web-based portal for project creation workflow
- **Automated Template Deployment**
 - Ability to auto-select and deploy project templates based on class
 - Ability to initialize folder structure (SharePoint, file repositories)
 - Ability to provision access control rules for folders/permissions

- Ability to configure Common Data Environment (CDE) structure
- Ability to deploy project-specific Revit templates
- Ability to deploy project-specific database templates
- Ability to configure team collaboration spaces (Teams, SharePoint)
- Ability to auto-generate Kanban boards for task management
- **Tool & Service Configuration**
 - Ability to provision project tools based on template configuration
 - Ability to configure approval workflows for project artifacts
 - Ability to establish project-level governance rules
 - Ability to link external tools and services via APIs (Autodesk ACC, Revit)
 - Ability to configure project standards and discipline-specific settings
 - Ability to seed project with reference data and best practices
- **Data Model Instantiation**
 - Ability to create basic project data model instance
 - Ability to initialize project context (project entity, disciplines, roles, geographical areas)
 - Ability to associate standards/guidelines with project
 - Ability to establish baseline master data for project

1.2 Project Deprovisioning & Archival

Core Capability: Systematic project closure and historical data preservation

- **Project Closeout Workflow**
 - Ability to mark project as closed/archived
 - Ability to trigger closeout checklist/procedures
 - Ability to validate all deliverables are complete
 - Ability to generate project closure reports
- **Data Archival & Preservation**
 - Ability to archive project data at project completion
 - Ability to maintain historical record for reference
 - Ability to compress/optimize archived data
 - Ability to maintain audit trail for archived content
 - Ability to enable read-only access to historical projects
 - Ability to extract lessons learned data for future use
- **Knowledge Capture at Closeout**
 - Ability to capture project-specific best practices
 - Ability to extract reusable templates/configurations
 - Ability to document lessons learned
 - Ability to update organizational knowledge base with project insights

SECTION 2: USER & ACCESS MANAGEMENT

2.1 User Onboarding & Project Membership

Core Capability: Streamline project team assembly and system access provisioning

- **User Addition to Projects**
 - Ability to add users to project team
 - Ability to assign user roles within project context
 - Ability to specify user discipline/specialization
 - Ability to determine user access level (viewer, contributor, admin)
 - Ability to configure time-bound project membership (start/end dates)
- **Automated Access Provisioning**
 - Ability to auto-grant access to project tools based on role
 - Ability to configure folder/repository access based on role and discipline
 - Ability to grant access to project data environments
 - Ability to provision access to collaborative spaces (Teams, SharePoint)
 - Ability to establish vendor-agnostic access interface
 - Ability to synchronize access across multiple systems/vendors
 - Ability to apply discipline-specific tool access rules
 - Ability to validate Azure AD/identity integration
- **Approval Workflows**
 - Ability to trigger approval workflows for access requests
 - Ability to route requests to project managers/approvers
 - Ability to enforce approval rules and SLAs
 - Ability to audit approval decisions
 - Ability to log access grant/deny events
- **User Profile & Context**
 - Ability to link user to department/organizational unit
 - Ability to capture user's home office/location
 - Ability to establish user's primary discipline
 - Ability to derive user permissions from organizational context
 - Ability to maintain user capability/skill registry

2.2 User Offboarding & Access Revocation

Core Capability: Systematic removal of user access and project departure

- **User Removal Process**
 - Ability to remove user from project team
 - Ability to log offboarding date and reason
 - Ability to trigger access revocation cascade
 - Ability to transfer ownership of user's artifacts/tasks
- **Access Revocation Across Systems**
 - Ability to revoke access from all project tools simultaneously
 - Ability to revoke folder/file permissions
 - Ability to close collaborative space access
 - Ability to maintain audit trail of revocations
 - Ability to verify complete access removal
- **Knowledge Transfer**
 - Ability to identify critical knowledge held by departing user
 - Ability to facilitate knowledge transfer workflows
 - Ability to reassign responsibility for artifacts/processes

2.3 Role & Responsibility Management

Core Capability: Define and maintain clear accountability structures

- **Role Definition & Assignment**
 - Ability to define project roles (PM, Designer, Engineer, Reviewer, etc.)
 - Ability to assign roles with specific permissions/capabilities
 - Ability to manage role-based access policies
 - Ability to link roles to approval authorities
 - Ability to establish responsibility matrices (RACI)
- **Responsibility Tracking**
 - Ability to establish "who does what" mapping
 - Ability to track responsibility assignments over time
 - Ability to identify single points of failure (single-accountable)
 - Ability to escalate responsibility when needed

SECTION 3: PROJECT PLANNING & DELIVERY MANAGEMENT

3.1 Work Structure Definition

Core Capability: Define and organize project work scope

- **Work Breakdown Structure (WBS)**
 - Ability to create hierarchical work structure
 - Ability to define work packages
 - Ability to decompose packages into tasks
 - Ability to establish parent-child relationships
 - Ability to assign effort/duration to work items
 - Ability to link work to disciplines/teams
- **Milestone & Task Management**
 - Ability to define project milestones
 - Ability to establish dependencies between tasks/milestones
 - Ability to define task start/end dates
 - Ability to establish critical path
 - Ability to identify schedule constraints
- **Deliverables Definition**
 - Ability to define deliverable types (documents, models, data, etc.)
 - Ability to link deliverables to work packages
 - Ability to establish deliverable acceptance criteria
 - Ability to maintain master document register
 - Ability to track deliverable status (not started, in progress, complete, delivered)

3.2 Project Monitoring & Observability

Core Capability: Continuous visibility into project health and execution

- **Progress Tracking**
 - Ability to capture actual progress vs. planned
 - Ability to calculate completion percentages
 - Ability to identify schedule variances
 - Ability to update progress automatically from linked tools
 - Ability to provide dashboard view of overall progress
- **Cost & Resource Tracking**

- Ability to track labor costs against budget
- Ability to track resource utilization
- Ability to identify cost overruns
- Ability to forecast final cost at completion
- Ability to provide cost visibility by discipline/team
- **Risk, Issue & Delay Detection**
 - Ability to identify and flag schedule delays
 - Ability to escalate delay notifications
 - Ability to track identified risks and mitigation status
 - Ability to log issues and track resolution
 - Ability to identify critical path impacts
 - Ability to highlight blockers preventing progress
- **Dashboard & Reporting**
 - Ability to display project health dashboard
 - Ability to show key metrics (progress %, cost variance, schedule variance)
 - Ability to visualize schedule timeline
 - Ability to highlight at-risk areas
 - Ability to provide drill-down to detail level
 - Ability to generate project status reports
 - Ability to support "view filtered by user context" functionality
- **Notification & Escalation**
 - Ability to alert stakeholders to issues/risks
 - Ability to escalate critical items
 - Ability to establish notification rules

SECTION 4: DATA QUALITY & VALIDATION

4.1 Data Classification & Organization

Core Capability: Systematic categorization and semantic organization of project data

- **Classification System Configuration**
 - Ability to define classification taxonomies
 - Ability to establish classification dimensions (type, discipline, status, etc.)
 - Ability to create discipline-specific classification schemes
 - Ability to support discipline-agnostic common classifications

- Ability to make classification configurable per GBA/business area
- Ability to establish cross-discipline classification consistency
- **Data Tagging & Metadata**
 - Ability to auto-tag data based on context
 - Ability to capture manual metadata entry
 - Ability to enforce metadata requirements
 - Ability to associate metadata with data objects
 - Ability to track metadata lineage and changes

4.2 Data Validation & Quality Rules

Core Capability: Enforce data quality standards and prevent non-compliance

- **Validation Rule Engine**
 - Ability to define data quality rules (by discipline, by data type)
 - Ability to validate data against standards at input
 - Ability to flag data quality issues
 - Ability to prevent storage of non-compliant data
 - Ability to run validation reports
 - Ability to track data quality metrics over time
- **Standards Enforcement**
 - Ability to define project standards/specifications
 - Ability to validate compliance against standards
 - Ability to embed validation in design tools (Revit, etc.)
 - Ability to support bsdd (buildingSMART Data Dictionary) validation
 - Ability to prevent error/non-compliance through tool integration
- **Data Quality Monitoring**
 - Ability to continuously monitor data quality
 - Ability to identify anomalies and outliers
 - Ability to generate data quality scorecards
 - Ability to track quality trends

4.3 Master Data & Reference Data Management

Core Capability: Centralized management of canonical reference information

- **Master Data Catalog**
 - Ability to maintain central registry of reference entities
 - Ability to define master data entities (materials, components, standards, etc.)

- Ability to maintain master data versions
- Ability to track master data changes and lineage
- **Reference Data Library**
 - Ability to create organizational reference database
 - Ability to store and version reference data
 - Ability to make reference data available for project reuse
 - Ability to extract best practices into reference library
 - Ability to manage reference data by discipline
 - Ability to include material specs, piping specs, fluids, standard components
- **Data Reuse & Cross-Project Leverage**
 - Ability to apply reference data to new projects
 - Ability to inherit from similar previous projects
 - Ability to parameterize designs based on reference data
 - Ability to enable "design by variant" approach
 - Ability to modify reference data for project-specific needs
 - Ability to track which projects use which reference data

SECTION 5: INFORMATION GOVERNANCE & KNOWLEDGE MANAGEMENT

5.1 Knowledge Capture & Representation

Core Capability: Systematic capture and organization of organizational knowledge

- **Process Documentation & Capture**
 - Ability to capture process workflows (visual, textual, video)
 - Ability to record screen captures of working processes
 - Ability to use AI to translate process videos into documented steps
 - Ability to maintain scalable knowledge capture (not manual 10-hour processes)
 - Ability to capture discipline-specific workflows
 - Ability to maintain version control for process documents
- **AI-Powered Knowledge Extraction**
 - Ability to extract knowledge from unstructured sources
 - Ability to unpack data from PDFs and closed containers
 - Ability to apply AI classification to extracted data
 - Ability to make extracted data LLM-friendly for AI tools

- Ability to create AI-consumable data structures from legacy documents
- Ability to support AI-powered rules checkers with structured data
- **Knowledge Base & Ontology**
 - Ability to organize knowledge hierarchically
 - Ability to establish semantic relationships between concepts
 - Ability to build discipline-specific ontologies
 - Ability to maintain organization-wide knowledge graph
 - Ability to support natural language search over knowledge base
 - Ability to make knowledge accessible to AI agents
- **Best Practices Repository**
 - Ability to capture and store best practices
 - Ability to classify best practices by discipline/domain
 - Ability to version and update best practices
 - Ability to track adoption and effectiveness of practices
 - Ability to extract best practices from completed projects

5.2 Metadata Management & Semantic Integration

Core Capability: Establish semantic consistency and data meaning across systems

- **Metadata Framework**
 - Ability to define metadata standards
 - Ability to establish metadata consistency rules
 - Ability to maintain metadata catalog
 - Ability to track metadata lineage
 - Ability to resolve metadata conflicts across sources
- **Data Context & Relationships**
 - Ability to establish "view filtered by user context"
 - Ability to generate context-specific data extracts
 - Ability to auto-trigger extraction when data changes
 - Ability to maintain different data facets for different purposes
 - Ability to support query-based and manifest/cache-based approaches
- **Semantic Web & Linked Data**
 - Ability to represent data as semantic triples (subject-predicate-object)
 - Ability to establish linked data across organizational silos
 - Ability to create uniform data access patterns

5.3 Data Extraction & Transformation

Core Capability: Convert project data into usable forms for different purposes

- **Data Extraction Services**
 - Ability to extract data from source systems
 - Ability to filter and transform extracted data
 - Ability to apply context to extracted data
 - Ability to generate multiple data facets from single source
 - Ability to query on-the-fly or pre-materialize extracts
 - Ability to refresh extracts on schedule or on-demand
- **Format Conversion & Interoperability**
 - Ability to convert between data formats (IFC, SQL, Excel, JSON, etc.)
 - Ability to support IFC as data container (with/without 3D)
 - Ability to validate converted data
 - Ability to maintain data integrity during conversion
 - Ability to support multiple export formats for different uses
- **Data Pipeline & ETL**
 - Ability to define data ingestion pipelines
 - Ability to transform data through pipeline stages
 - Ability to apply quality checks at pipeline stages
 - Ability to log data lineage through transformations
 - Ability to handle streaming and batch processing

SECTION 6: AI INTEGRATION & INTELLIGENCE CAPABILITIES

6.1 AI-Powered Rules & Compliance Checking

Core Capability: Automated intelligent validation and compliance verification

- **Rules Engine Integration**
 - Ability to define business rules (by discipline, by domain)
 - Ability to execute rules against project data
 - Ability to flag rule violations
 - Ability to provide remediation recommendations
 - Ability to support AI Rules Checker applications

- Ability to embed rule checking in design tools
- **AI-Powered Validation**
 - Ability to apply machine learning models to detect anomalies
 - Ability to classify data using trained models
 - Ability to predict issues based on historical patterns
 - Ability to provide confidence scores for predictions
 - Ability to continuously improve models with new data

6.2 AI-Powered Code Generation & Automation

Core Capability: Leverage AI for development and task automation

- **Vibe Coding & AI-Assisted Development**
 - Ability to support AI-assisted code generation (GitHub Copilot, Claude, etc.)
 - Ability to build prototypes rapidly using AI
 - Ability to generate boilerplate code/templates
 - Ability to auto-generate documentation from code
 - Ability to support Replit-based full-stack application development
- **Automation & RPA**
 - Ability to automate repetitive workflows
 - Ability to execute automation scripts triggered by events
 - Ability to build complex automation using AI agents
 - Ability to maintain audit trail of automated actions

6.3 AI Agents & Intelligent Assistants

Core Capability: Deploy specialized AI agents for domain-specific tasks

- **3D Model Query & Analysis**
 - Ability to query 3D models using natural language
 - Ability to extract information from 3D models
 - Ability to perform geometric analysis
 - Ability to identify conflicts/clashes in models
 - Ability to support AI agents talking to Revit/ACC via APIs
- **Domain-Specific AI Agents**
 - Ability to develop specialized agents for each workflow
 - Ability to train agents on discipline-specific knowledge
 - Ability to operate agents directly within production tools
 - Ability to support competition-winning workflow agents (e.g., Henning Larssen approach)

- **Chatbot & Conversational Interface**

- Ability to provide natural language interface to data
- Ability to answer questions about project data
- Ability to support chatbots for Revit models
- Ability to enable semantic search and discovery

6.4 AI Implementation Strategy

Core Capability: Manage both top-down and bottom-up AI adoption

- **Top-Down AI Enablement**

- Ability to enable vendor out-of-box AI (Autodesk, Tier 1/2 vendors)
- Ability to integrate vendor AI capabilities
- Ability to establish enterprise AI standards

- **Bottom-Up AI Enablement**

- Ability to support grassroots vibe coding on projects
- Ability to maintain database of vibe-coded solutions
- Ability to provide recommended tech stacks for AI solutions
- Ability to establish AI solution patterns/templates
- Ability to share learnings across projects

SECTION 7: SYSTEM INTEGRATION & INTEROPERABILITY

7.1 API-Based Integration Framework

Core Capability: Seamless connection between UPE and enterprise systems

- **Internal System Integration**

- Ability to integrate with CRM (customer data, projects, opportunities)
- Ability to integrate with ERP/Maconomy (financial data, time reporting, billing)
- Ability to integrate with Microsoft suite (Teams, SharePoint, Power BI, Planner, Azure)
- Ability to integrate with Service Now (ITSM, IT services)
- Ability to integrate with Workday (HR, organizational hierarchy)
- Ability to sync organizational structure and user data

- **Design Tool Integration**

- Ability to integrate with Autodesk Revit via APIs

- Ability to integrate with Autodesk Forma/ACC via APIs
- Ability to support other CAD tools and BIM platforms
- Ability to pull data from design tools
- Ability to validate designs against project rules
- Ability to push updates/corrections back to tools
- **Data Environment Integration**
 - Ability to connect to Common Data Environments (CDE)
 - Ability to sync with GIS platforms
 - Ability to integrate with visualization tools
 - Ability to support multiple CDE vendors
- **Unified Abstraction Layer**
 - Ability to provide vendor-agnostic APIs
 - Ability to abstract vendor-specific differences
 - Ability to provide consistent interface regardless of underlying vendor
 - Ability to route requests to appropriate vendor systems
 - Ability to add new vendors without breaking existing integrations

7.2 Connector & Adapter Development

Core Capability: Extend integration capabilities systematically

- **Connector Architecture**
 - Ability to develop new connectors for additional systems
 - Ability to standardize connector patterns
 - Ability to version and manage connectors
 - Ability to test connectors before deployment
- **Data Adapter Patterns**
 - Ability to create adapters for different data formats
 - Ability to establish transformation patterns
 - Ability to maintain adapter library

7.3 Vendor Independence & Interoperability

Core Capability: Avoid vendor lock-in and maintain flexibility

- **Open Standards Support**
 - Ability to support Open BIM standards
 - Ability to use open formats for data exchange
 - Ability to implement W3C/OASIS standards

- Ability to maintain vendor-agnostic architecture
- **Multi-Vendor Support**
 - Ability to support multiple vendors for same capability
 - Ability to switch vendors without data loss
 - Ability to maintain data portability
 - Ability to avoid "all eggs in one basket" scenarios

SECTION 8: EMBEDDED PROCESS AUTOMATION & WORKFLOW INTEGRATION

8.1 Process Embedding in Tools

Core Capability: Bake processes directly into design and collaboration tools

- **Interdisciplinary Collaboration**
 - Ability to enable cross-discipline teams in single environment
 - Ability to establish discipline-specific views of shared data
 - Ability to support concurrent work across disciplines
 - Ability to manage dependencies between disciplines
 - Ability to establish communication patterns between disciplines
- **Error Prevention & Non-Compliance Avoidance**
 - Ability to enforce process rules within tools
 - Ability to prevent invalid state transitions
 - Ability to block non-compliant actions
 - Ability to provide contextual guidance
 - Ability to suggest corrections
- **Simplified Designer Workflows**
 - Ability to streamline engineer/designer workflows
 - Ability to reduce non-value-add activities
 - Ability to auto-populate standard information
 - Ability to provide templates for common patterns
 - Ability to auto-generate documentation

8.2 Approval & Governance Workflows

Core Capability: Manage change, reviews, and approvals systematically

- **Review & Approval Processes**
 - Ability to define review workflows
 - Ability to route work for review/approval
 - Ability to establish reviewer queues
 - Ability to track approval status
 - Ability to manage concurrent vs. serial reviews
 - Ability to enforce sign-off requirements
- **Change Management**
 - Ability to track changes to project data
 - Ability to maintain change history
 - Ability to enable rollback to previous versions
 - Ability to log change authorship and timestamps

8.3 Event-Driven Automation

Core Capability: Trigger automated actions based on system events

- **Event Detection**
 - Ability to detect data changes
 - Ability to trigger workflows on events
 - Ability to establish event rules
 - Ability to cascade events through related entities
- **Automated Actions**
 - Ability to execute automation in response to events
 - Ability to maintain audit trail
 - Ability to support notifications and escalations

SECTION 9: USER EXPERIENCE & INTERFACE DESIGN

9.1 Project Portal & Navigation

Core Capability: Provide intuitive entry point and navigation

- **Project Landing Page**
 - Ability to display project overview
 - Ability to show quick stats and key info

- Ability to provide access to main project functions
- Ability to highlight recent activities
- Ability to show notifications/alerts
- **Navigation & Discovery**
 - Ability to browse project portfolio
 - Ability to filter projects by status, type, GBA, etc.
 - Ability to search across projects
 - Ability to provide contextual navigation
 - Ability to support "most frequently used" shortcuts

9.2 Dashboards & Reporting

Core Capability: Provide visibility and insights through tailored views

- **Project Dashboard**
 - Ability to display project-specific metrics
 - Ability to provide drill-down capability
 - Ability to support custom dashboards
 - Ability to show progress, cost, risks, issues
- **Executive Dashboard**
 - Ability to aggregate data from multiple projects
 - Ability to show portfolio-level metrics
 - Ability to support portfolio-level reporting
 - Ability to identify at-risk projects
- **Team Dashboard**
 - Ability to show team workload
 - Ability to display assigned tasks
 - Ability to show calendar/schedule
 - Ability to highlight blockers/dependencies

9.3 Multi-Project & Multi-Context Support

Core Capability: Support users working across multiple projects

- **Project Context Switching**
 - Ability to switch between projects rapidly
 - Ability to maintain context when switching
 - Ability to show active project indicator
 - Ability to preserve user settings per project

- **Unified Task List**

- Ability to aggregate tasks across all projects
- Ability to filter by project, priority, deadline
- Ability to provide consolidated view (like MS Planner)
- Ability to track workload across projects

SECTION 10: FOUNDATIONAL REQUIREMENTS & TECHNICAL ENABLEMENT

10.1 Data Model & Semantic Foundation

Core Capability: Establish canonical project context and relationships

- **Project Core Data Model**

- Ability to define and maintain project entity
- Ability to define and maintain discipline entities
- Ability to define and maintain role/responsibility entities
- Ability to establish geographical areas/scopes
- Ability to link standards/specifications to projects
- Ability to establish project-to-organization relationships
- Ability to support discipline-specific data models alongside core model

- **Data Consistency & Integration**

- Ability to maintain single source of truth for project context
- Ability to synchronize derived data from source systems
- Ability to resolve conflicts in distributed data
- Ability to establish data lineage and dependencies

10.2 Configuration & Customization

Core Capability: Enable adaptation to different business needs

- **Template System**

- Ability to create and manage project templates
- Ability to establish template versioning
- Ability to apply templates to new projects
- Ability to customize templates for GBA/domain specifics

- **Configuration Management**

- Ability to configure system behavior per GBA
- Ability to establish discipline-specific rules
- Ability to enable feature flags/toggles
- Ability to maintain configuration versions

10.3 Scalability & Performance

Core Capability: Support growth and concurrent usage

- **Scalable Architecture**
 - Ability to support projects of varying sizes
 - Ability to handle large datasets efficiently
 - Ability to support real-time and batch processing
 - Ability to scale horizontally for increased load
- **Cloud Infrastructure**
 - Ability to leverage Microsoft Azure
 - Ability to scale cloud resources dynamically
 - Ability to establish disaster recovery
 - Ability to maintain high availability

10.4 Security & Access Control

Core Capability: Protect data and control access

- **Authentication & Authorization**
 - Ability to integrate with Azure AD
 - Ability to establish role-based access control (RBAC)
 - Ability to enforce granular permissions
 - Ability to manage API access and tokens
 - Ability to support OAuth/OIDC patterns
- **Data Protection**
 - Ability to encrypt data at rest and in transit
 - Ability to mask sensitive data
 - Ability to establish audit trails
 - Ability to comply with security standards
- **Compliance & Governance**
 - Ability to maintain audit logs
 - Ability to support compliance reporting
 - Ability to establish data retention policies

- Ability to support right-to-be-forgotten

10.5 Documentation & Knowledge Management

Core Capability: Enable system learning and communication

- **Architecture Documentation**
 - Ability to document system design decisions
 - Ability to maintain architectural diagrams
 - Ability to establish design patterns library
 - Ability to document technical standards
- **User Documentation**
 - Ability to provide user guides and help
 - Ability to maintain FAQ database
 - Ability to support contextual help
 - Ability to provide video tutorials
- **Change & Release Management**
 - Ability to document releases and changes
 - Ability to track breaking changes
 - Ability to communicate to users
 - Ability to provide migration guides

SECTION 11: PLATFORM GOVERNANCE & ROADMAP MANAGEMENT

11.1 Capability Roadmap & Release Planning

Core Capability: Manage platform evolution systematically

- **Feature Prioritization**
 - Ability to assess feature impact and effort
 - Ability to prioritize based on business value
 - Ability to manage competing requirements
 - Ability to establish release schedule
- **Release Planning**
 - Ability to plan releases (quarterly, annual)
 - Ability to manage dependencies between features

- Ability to track feature status and progress
- Ability to communicate roadmap to stakeholders
- **Portfolio Governance**
 - Ability to track multiple product initiatives
 - Ability to identify feature/capability dependencies
 - Ability to ensure coherent evolution
 - Ability to support tool like Aha! for roadmapping

11.2 GBA Requirements Aggregation

Core Capability: Gather and synthesize requirements across business units

- **Requirements Capture**
 - Ability to capture requirements from each GBA
 - Ability to standardize requirement format
 - Ability to establish common vs. specific requirements
 - Ability to identify cross-GBA dependencies
- **Requirements Analysis**
 - Ability to identify common capabilities across GBAs
 - Ability to establish reusability metrics
 - Ability to recommend standardization opportunities
 - Ability to highlight conflicting requirements

SECTION 12: MONITORING, DIAGNOSTICS & OPERATIONAL SUPPORT

12.1 System Health & Performance Monitoring

Core Capability: Understand and optimize system behavior

- **Performance Metrics**
 - Ability to monitor API response times
 - Ability to track system uptime/availability
 - Ability to monitor resource utilization
 - Ability to establish SLA compliance
- **Usage Analytics**
 - Ability to track feature usage

- Ability to identify adoption rates
- Ability to understand user workflows
- Ability to optimize based on usage patterns

12.2 Alerting & Incident Management

Core Capability: Respond to issues systematically

- **Alert Generation**
 - Ability to alert on SLA violations
 - Ability to alert on data quality issues
 - Ability to alert on security events
 - Ability to escalate critical alerts
- **Incident Tracking**
 - Ability to log incidents
 - Ability to track resolution status
 - Ability to maintain incident history
 - Ability to perform root cause analysis

SECTION 13: BUILD VS. BUY & TECHNOLOGY ENABLEMENT

13.1 Technology Stack Recommendations

Core Capability: Select and integrate optimal platform components

- **Approved Technology Components**
 - Microsoft Teams/SharePoint (collaboration, storage)
 - Microsoft Power BI (analytics, dashboards)
 - Microsoft Power Automate (workflow automation)
 - Microsoft Planner (task management)
 - Microsoft Azure (cloud infrastructure)
 - Autodesk APIs (design tool integration)
 - Third-party integrations (CRM, ERP, etc.)
- **Build vs. Buy Decision Framework**
 - Ability to evaluate build vs. buy for each component
 - Ability to assess integration complexity

- Ability to manage costs vs. capability trade-offs
- Ability to reduce maintenance burden through vendor solutions

13.2 Enabler Development & Tooling

Core Capability: Build foundational components enabling faster development

- **Data Model Development**
 - Ability to rapidly iterate on data model
 - Ability to establish canonical schema
 - Ability to document data model
 - Ability to generate code from schema
- **Template & Script Library**
 - Ability to create reusable templates
 - Ability to develop automation scripts
 - Ability to establish pattern library
 - Ability to version and share across teams
- **Connector & Integration Enablers**
 - Ability to develop connector templates
 - Ability to establish adapter patterns
 - Ability to enable teams to build connectors
 - Ability to test and validate connectors

SECTION 14: SPECIAL CAPABILITY DOMAINS

14.1 BIM/GIS & Visualization Capabilities

Core Capability: Support geometry-based project data

- **3D Model Management**
 - Ability to manage Revit models
 - Ability to establish model storage/repository
 - Ability to track model versions
 - Ability to support multi-discipline coordination
- **GIS Integration**
 - Ability to integrate geographic data
 - Ability to establish location context for projects

- Ability to use GIS for site visualization
- Ability to overlay project data on maps

14.2 Time Reporting & Resource Management

Core Capability: Track labor and resource allocation

- **Time & Resource Tracking**
 - Ability to integrate with time reporting systems
 - Ability to track resource allocation to projects/tasks
 - Ability to identify resource conflicts
 - Ability to forecast resource needs
 - Ability to support billing integration

14.3 Contracting & Legal Compliance

Core Capability: Manage project contracts and obligations

- **Contract Management**
 - Ability to track project contracts
 - Ability to maintain contract compliance
 - Ability to track contract deliverables
 - Ability to manage contract changes/amendments
 - Ability to support billing from contracts

14.4 Document Generation & Delivery Management

Core Capability: Automate document production and delivery

- **Automated Documentation**
 - Ability to auto-generate documents from data
 - Ability to establish document templates
 - Ability to populate templates from project data
 - Ability to maintain document versions
 - Ability to package and deliver documents
- **Master Document Register**
 - Ability to establish document register
 - Ability to track document versions
 - Ability to specify deliverable documents
 - Ability to automate document identification

- Ability to manage document status

CROSS-CUTTING CONCERNS & HORIZONTAL CAPABILITIES

C1: Data Lineage & Impact Analysis

- **Capability:** Track data origins and understand downstream impacts
 - Ability to establish data lineage from source to use
 - Ability to identify data dependencies
 - Ability to perform impact analysis on data changes
 - Ability to understand which systems depend on data
 - Ability to trace issues to root causes

C2: Audit & Compliance

- **Capability:** Maintain comprehensive audit trail and regulatory compliance
 - Ability to log all significant system actions
 - Ability to maintain audit trail for compliance
 - Ability to support compliance reporting
 - Ability to establish retention policies
 - Ability to support data subject access requests

C3: Notifications & Communication

- **Capability:** Keep users informed and engaged
 - Ability to send notifications for relevant events
 - Ability to establish notification preferences
 - Ability to avoid notification fatigue
 - Ability to escalate critical communications
 - Ability to support in-app and email notifications

C4: Search & Discovery

- **Capability:** Help users find information and capabilities
 - Ability to search across project data
 - Ability to support natural language search

- Ability to index documents and metadata
- Ability to provide relevance ranking
- Ability to discover related data

C5: Extensibility & Plugin Architecture

- **Capability:** Allow customization without modifying core
 - Ability to support third-party extensions
 - Ability to establish plugin standards
 - Ability to sandbox plugins safely
 - Ability to manage plugin lifecycle
 - Ability to support developer ecosystem

FUNCTIONAL BLOCKS SUMMARY TABLE

Functional Block	Core Responsibilities	Key Capabilities Count
Project Lifecycle Management	Initialization, operation, closeout, archival	15+
User & Access Management	Onboarding, offboarding, role/responsibility	20+
Planning & Delivery	WBS, milestones, deliverables, progress tracking	18+
Data Quality & Governance	Classification, validation, master data, reference data	15+
Knowledge Management	Capture, representation, extraction, ontology	14+
AI Integration	Rules, agents, automation, intelligence	14+
System Integration	APIs, connectors, vendors, standards	12+
Process Automation	Embedding, workflows, approval, events	10+
User Experience	Portal, dashboards, context, navigation	10+

Functional Block	Core Responsibilities	Key Capabilities Count
Foundations	Data model, configuration, scalability, security	14+
Governance & Roadmap	Prioritization, GBA aggregation, release management	8+
Operations & Monitoring	Health, performance, alerts, incidents	8+
Technology Enablement	Stack, build/buy, enablers, tools	8+
Special Domains	BIM/GIS, time reporting, contracts, documentation	10+
Cross-Cutting	Lineage, audit, notifications, search, extensibility	10+

Total Functional Capabilities Identified: 175+

METHODOLOGY & NEXT STEPS

Document Scope

This skeleton represents a comprehensive pool of functional blocks extracted from:

- [Brainstorming.md](#) requirements document
- Three project launch series meeting transcripts
- Architect chat discussions
- Real-world precedents (Mott, Arup, Henning Larssen)
- Technology partnerships (Autodesk, Microsoft)

Intentional Inclusion of Variations

The document intentionally includes:

- **Duplicative capabilities** (different wordings of same function)
- **Complementary approaches** (multiple ways to achieve same outcome)
- **Potentially conflicting options** (e.g., query-based vs. cached data)

- **Reach capabilities** (aspirational but discussed features)

Next Phases

Phase 1 (Current): Skeleton & functional pool definition ✓

Phase 2: Detailed requirements specification

- Expand each capability with acceptance criteria
- Define dependencies between capabilities
- Establish priority & sequencing
- Identify conflicts and trade-offs

Phase 3: Technical architecture mapping

- Map functional blocks to technical components
- Define APIs and data contracts
- Establish integration patterns
- Create deployment architecture

Phase 4: Implementation roadmap

- Prioritize functional blocks by value/effort
- Establish release phases
- Allocate teams and resources
- Set timelines and milestones

Document Status: Draft Skeleton - Ready for Review & Refinement

Last Updated: [Generation Date]

Next Review: Post-stakeholder feedback cycle